

ASEN-6062

HWK #1

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1) Poilard ch. 1

3.3) given: $\vec{a} = \vec{r}, \vec{b} = \vec{v}$
 $(\vec{a} \cdot \vec{b})^2 + (\vec{a} \times \vec{b})^2 = a^2 b^2$

Show: $v^2 = \dot{r}^2 + \frac{c^2}{r^2}$

$$(\vec{r} \cdot \vec{v})^2 + (\vec{r} \times \vec{v})^2 = r^2 v^2$$

$$(\vec{r} \cdot \vec{v})^2 + \vec{c} \cdot \vec{c} = r^2 v^2$$

$$(\vec{r} \cdot \dot{\vec{r}})^2 + c^2 = r^2 v^2$$

$$r^2 \dot{r}^2 + c^2 = r^2 v^2$$

$$v^2 = \dot{r}^2 + \frac{c^2}{r^2}$$

In physical terms, $\dot{r} = \text{radial velocity}$
 $\frac{c^2}{r^2} = \text{tangential velocity}$

$$T = 0 + h \quad \frac{1}{2} v^2 = f_1(r) + h$$

$$\frac{1}{2} (r \dot{r}^2 + \frac{c^2}{r^2}) = f_1(r) + h$$

$$\frac{1}{2} (r \dot{r}^2 + c^2) = r^2 [f_1(r) + h]$$

$$r \dot{r}^2 + c^2 = 2r^2 [f_1(r) + h]$$

4.1) show: if $0 < e < 1$ or $e > 1$

$$Ma |e^2 - 1| = c^2$$

$$a = \frac{r_p + r_a}{2}$$

$$a = \frac{r_p + r_a}{2}$$

$$r = \frac{a^2/\mu}{1 + e \cos f} = e \left(\frac{c^2}{\mu e} - r \cos f \right)$$

$$r_p = \frac{c^2/\mu}{1+e} \quad r_a = \frac{c^2/\mu}{1-e}$$

$$a = \frac{1}{2} \left[\frac{c^2/\mu}{1-e} + \frac{c^2/\mu}{-(1+e)} \right]$$

$$2a\mu = \frac{-c^2(1+e+1-e)}{(1-e)(1+e)}$$

$$2a\mu = -\frac{2c^2}{1-e^2}$$

$$\boxed{a\mu(c^2-1) = c^2}$$

5.3) given: $A_{ell} = \pi a^2 (1-e^2)^{1/2}$ $\dot{A} = c/2$ if $c \neq 0$

show: if $\underbrace{0 < e < 1}_{\text{ellipse}}$ $P = \left(\frac{2\pi}{\sqrt{\mu}} \right) a^{3/2}$

$$A = c/2 P$$

$$P = \frac{2A}{c} = \frac{2\pi a^2 (1-e^2)^{1/2}}{c}$$

$$= \frac{2\pi a^2 (1-e^2)^{1/2}}{\sqrt{\mu a |1-e^2|}}$$

$$= \frac{2\pi a^{3/2}}{\sqrt{\mu}}$$

$$c^2 = \mu a |e^2 - 1|$$

$$c^2 = \mu a |-(1-e^2)|$$

$$\boxed{P = \frac{2\pi a^{3/2}}{\sqrt{\mu}}}$$

5.4) define moment of inertia

$$2I = mr^2$$

prove $\ddot{I} = 2\dot{T} - \dot{U} = \dot{T} + \dot{h} = \dot{U} + 2\dot{h}$

$$I = \frac{1}{2} mr^2 = \frac{1}{2} m \vec{r} \cdot \vec{r}$$

$$\begin{aligned} I &= \frac{1}{2} m r^2 = \frac{1}{2} m \vec{r} \cdot \vec{r} \\ \dot{I} &= \frac{1}{2} m (\vec{r} \cdot \dot{\vec{r}} + \dot{\vec{r}} \cdot \vec{r}) \\ &= m \cdot \vec{r} \cdot \dot{\vec{r}} \end{aligned}$$

$$\begin{aligned} \ddot{I} &= m (\dot{\vec{r}} \cdot \dot{\vec{r}} + \vec{r} \cdot \ddot{\vec{r}}) \\ &= m (\dot{r}^2 + \vec{r} \cdot \ddot{\vec{r}}) \quad \ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r} \\ &= \underbrace{m v^2}_{2T} + \vec{r} \cdot m \left(-\frac{\mu}{r^3} \vec{r} \right) \\ &= 2T + \underbrace{m \left(-\frac{\mu}{r^3} r^2 \right)}_U \end{aligned}$$

$$\begin{aligned} \ddot{I} &= 2T - U = 2T - (T - h) \\ &= 2U + 2h - U = \end{aligned}$$

$$\boxed{\ddot{I} = U + 2h = T + h}$$

Q.3) given: u = eccentric anomaly
 f = true anomaly

$$r = a(e \cos u - 1) \quad (8.2) \quad h > 0$$

$$n(t - \tau) = e \sinh u - u$$

$$r = a(1 - e \cos u)$$

(mean anomaly) $n(t - \tau) = u - e \sin u \quad (8.3) \quad h < 0$

$$r = \frac{a(1 - e^2)}{1 + e \cos f} \quad (8.5)$$

$h < 0$ (elliptical)

trig identity: $\tan^2 \frac{u}{2} = \frac{1 - \cos u}{1 + \cos u}$

$$\frac{a(1 - e^2)}{1 + e \cos f} = a(1 - e \cos u)$$

$$1 - e^2 = (1 - e \cos u)(1 + e \cos f)$$

$$1 - e^2 = 1 + e \cos f - e \cos u - e^2 \cos u \cos f$$

$$r = \frac{a(1 - e^2)}{1 + e \cos f} \quad \left. \vphantom{\frac{a(1 - e^2)}{1 + e \cos f}} \right\} \text{Conic eqn}$$

$$a \cos u = ae + r \cos f$$

$$a \cos u = ae + \frac{a(1 - e^2) \cos f}{1 + e \cos f}$$

$$\begin{aligned} \cos u &= e(1 + e \cos f) + (1 - e^2) \cos f \\ &= \frac{e + \cos f}{1 + e \cos f} \end{aligned}$$

$$\tan^2 \frac{u}{2} = \frac{1 - \cos u}{1 + \cos u} = 1 - \frac{e + \cos f}{1 + e \cos f}$$

$$\begin{aligned} &= \frac{1 + e \cos f - e - \cos f}{1 + e \cos f} \\ &= \frac{1 + e \cos f + e + \cos f}{1 + e \cos f} \end{aligned}$$

$$= \frac{1 + e \cos f - e - \cos f}{1 + e \cos f + e + \cos f}$$

$$\tan^2 f = \frac{1 - \cos f}{1 + \cos f}$$

$$\begin{aligned}\tan^2 u &= \frac{(1 - \cos f) + e(\cos f - 1)}{(1 + \cos f) + e(\cos f + 1)} \\ &= \frac{(1 - \cos f)(1 - e)}{(1 + \cos f)(1 + e)} \\ &= \tan^2 \frac{f}{2}\end{aligned}$$

$$\tan^2 \frac{u}{2} = \frac{1 - e}{1 + e} \tan^2 \frac{f}{2}$$

$$\boxed{\tan \frac{f}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{u}{2}; h < 0}$$

for $h > 0$ (hyperbolic)

use same process but for $\cosh / \tanh$

$$\boxed{\Rightarrow \tan \frac{f}{2} = \sqrt{\frac{1+e}{1-e}} \tanh \frac{u}{2}, h > 0}$$

8.4 show solutions are unique

$0 < e \leq 1$ elliptical

$$\underbrace{n(t-T)} = u - e \sin u$$

Mean anomaly

$$f(u) = u - e \sin u$$

$$f'(u) = 1 - e \cos u \quad \cos u \in [-1, 1]$$

$$e \cos u|_{\max} = e(1) \leq 1$$

$f'(u) = 1 - e \geq 0 \quad \forall u \Rightarrow$ function is monotonically increasing so only 1 solution for given t .

$e \geq 1$

$$f(u) = e \sinh u - u$$

$$f'(u) = e \cosh u - 1 \quad \cosh(0) = 1 \quad \cosh u > 1 \quad \forall u \neq 0$$

$$e \cosh u \geq \cosh u \geq 1$$

$e \cosh u - 1 \geq 0 \quad \forall u \Rightarrow$ function is monotonically increasing

$$10.1 \quad (10.2) \quad l = u - e \sin u \quad 0 < e < 1$$

$$(10.1) \quad r = a(1 - e \cos u)$$

$$u - l = -e \sin u$$

periodic: $u(l) - l$
(eccentric anomaly - mean anomaly)

$$u - l = e \sin u$$

$$u - (l + 2\pi) = e \sin u$$

$$u - l = e \sin u + 2\pi = e \sin u$$

$$\text{odd: } u(-l) = -u(l)$$

$$-l = -(u - e \sin u)$$

$$-l = -(u - e \sin u)$$

$$u - l = -e \sin u$$

vanishes: $l = 0, l = \pi$

$$u - 0 = e \sin u$$

$$u = e \sin u \Rightarrow u = 0$$

$$u - \pi = e \sin u$$

$$u = e \sin u + \pi \Rightarrow u = \pi$$

cont. derivative

$$\frac{dl}{du} = \frac{d}{du}(u - e \sin u)$$

$$1 = \frac{du}{du} - e \cos u \frac{du}{du}$$

$$dl = (1 - e \cos u) du$$

$$\frac{du}{dl} = \frac{1}{1 - e \cos u}$$

$$\frac{d(u-l)}{dl} = \frac{du}{dl} - 1 = \frac{1}{1 - e \cos u} - 1$$

$$= \frac{1 - (1 - e \cos u)}{1 - e \cos u}$$

$$\frac{d(u-l)}{dl} = \frac{e \cos u}{1 - e \cos u} \rightarrow \text{continuous}$$

continuous & $\neq 0$

$$u(l) - l = \sum_{n=1}^{\infty} u_n \sin nl$$

$$\text{Prove: } \frac{2}{\pi n} \int_0^{\pi} \cos n(u - e \sin u) du$$

fourier coefficients:

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx$$

$$u_n = \frac{2}{\pi} \int_0^{\pi} (u(l) - l) (\sin nl) dl$$

(integration by parts)

$$f = u(l) - l$$

$$f' = \frac{du}{dl} - 1$$

$$\frac{dl}{du} = 1 - e \cos u$$

$$\frac{du}{dl} = \frac{1}{1 - e \cos u}$$

$$g = -\frac{1}{n} \cos nl$$

$$g' = \sin n l$$

$$\frac{du}{du} = 1 - e \cos u$$

$$\frac{du}{dl} = \frac{1}{1 - e \cos u}$$

$$f' = \frac{1}{1 - e \cos u} - 1$$

$$f' = \frac{e \cos u}{1 - e \cos u}$$

$$f' dl = \frac{e \cos u}{1 - e \cos u} (1 - e \cos u) du$$

$$f' dl = e \cos u du$$

$$u_n = \frac{2}{\pi n} \int_0^\pi \cos n l f'$$

$$= \frac{2}{\pi n} \int_0^\pi \cos n l e \cos u du$$

$$= \frac{2}{\pi n} \int_0^\pi \frac{\cos(u - e \sin u)}{e \cos u} e \cos u du$$

$$u_n = \frac{2}{\pi n} \int_0^\pi \cos(u - e \sin u) du$$

10.2 given: Q_0, Q_1 pts on elliptical orbit
 u_0, u_1 eccentric anomalies
 $u_1 > u_0$

prove: distance $Q_0 Q_1$ is $2a \sin \alpha \sin \beta$

$$\alpha = \frac{1}{2}(u_1 - u_0) \quad \cos \beta = e \cos \frac{1}{2}(u_1 + u_0)$$

$$0 < \beta < \pi \quad \gamma = \frac{1}{2}(u_1 + u_0)$$

$$\vec{r} = (a \cos u - ae, a \sqrt{1 - e^2} \sin u)$$

$$|\vec{r}_1 - \vec{r}_0|^2 = a^2 (\cos u_1 - \cos u_0)^2 - a^2 (1 - e^2) (\sin u_1 - \sin u_0)^2$$

$$(\cos u_1 - \cos u_0)^2 = \cos^2 u_1 - 2 \cos u_0 \cos u_1 + \cos^2 u_0$$

$$(\sin u_1 - \sin u_0)^2 = \sin^2 u_1 - 2 \sin u_0 \sin u_1 + \sin^2 u_0$$

$$\begin{aligned} \frac{|\vec{r}_1 - \vec{r}_0|^2}{a^2} &= \cos^2 u_1 + \cos^2 u_0 - 2 \cos u_0 \cos u_1 \\ &\quad - (1 - e^2) (\sin^2 u_1 + \sin^2 u_0 - 2 \sin u_0 \sin u_1) \end{aligned}$$

$$\cos u_1 - \cos u_0 = -2 \sin \frac{u_1 + u_0}{2} \sin \frac{u_1 - u_0}{2}$$

$$\sin u_1 - \sin u_0 = 2 \cos \frac{u_1 + u_0}{2} \sin \frac{u_1 - u_0}{2}$$

$$\frac{|\vec{r}_1 - \vec{r}_0|^2}{a^2} = 4 \sin^2 \alpha \sin^2 \gamma + 4(1 - e^2) \cos^2 \gamma \sin^2 \alpha$$

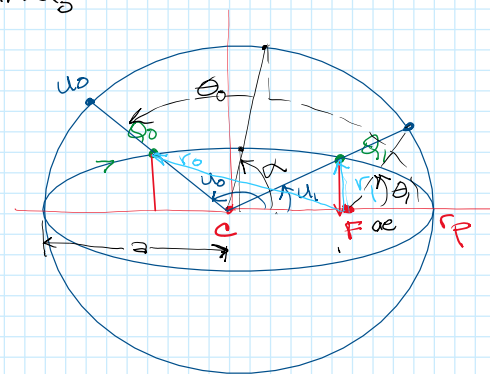
$$= 4 \sin^2 \alpha (\sin^2 \gamma + (1 - e^2) \cos^2 \gamma)$$

$$= 4 \sin^2 \alpha (\sin^2 \gamma + \cos^2 \gamma - e^2 \cos^2 \gamma)$$

$$= 4 \sin^2 \alpha (1 - e^2 \cos^2 \frac{u_1 + u_0}{2})$$

$$|\vec{r}_1 - \vec{r}_0|^2 = 4 \sin^2 \alpha \sin^2 \beta \cos^2 \beta$$

$$Q_0 Q_1 = 2 \sin \alpha \sin \beta$$



$$4.2 \quad \ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r} \quad (14.2)$$

$$(r\dot{r})^2 + \vec{c}^2 = 2(\mu r + h r^2) \quad c=0 \quad \mu = G(m_1 + m_2)$$

$$(r\dot{r})^2 = 2(\mu + h r) \quad r \rightarrow 0 \quad r\dot{r}^2 \rightarrow 2\mu$$

$$t \rightarrow t_1$$

Show: $|r\dot{r}|^{1/2} \rightarrow \sqrt{2\mu}$

$$r|t-t_1| \rightarrow (9/2\mu)^{1/3} \text{ as } t \rightarrow t_1$$

$$r\dot{r}^2 = 2\mu + 2hr^2$$

$$\sqrt{r\dot{r}^2} \rightarrow \sqrt{2\mu}$$

$$|r\dot{r}|^{1/2} \rightarrow \sqrt{2\mu} \quad \checkmark$$

$$\frac{dr}{dt} = \sqrt{\frac{2\mu}{r}}$$

$$\int r^{1/2} dr = \int \sqrt{2\mu} dt$$

$$\frac{2}{3} r^{3/2} = \sqrt{2\mu} t \Big|_{t_1}^{t_2}$$

$$\frac{2}{3} r^{3/2} = \sqrt{2\mu} (t - t_1)$$

$$r^{3/2} = \frac{3}{2} \sqrt{2\mu} (t - t_1)$$

$$r = \left[\frac{3}{2} \sqrt{2\mu} (t - t_1) \right]^{2/3}$$

$$r = \left(\frac{9}{2} \mu \right)^{1/3} (t - t_1)^{2/3}$$

$$\frac{r}{(t-t_1)^{2/3}} = \left(\frac{9}{2} \mu \right)^{1/3}$$

$$r(t-t_1)^{-2/3} \rightarrow \left(\frac{9}{2} \mu \right)^{1/3} \quad t \rightarrow t_1$$

$$16.1 \quad \ddot{\vec{r}} = \frac{F_c}{c} \vec{c} + F_r \frac{\vec{r}}{r} + F_a \frac{\vec{\alpha}}{\alpha}$$

$$16.2 \quad \dot{\vec{c}} = \vec{r} \times \left(-\frac{\mu}{r^3} \vec{r} + \ddot{\vec{r}} \right) = \vec{r} \times \ddot{\vec{r}}$$

$$\dot{\vec{c}} = \vec{r} \times \frac{F_c}{c} \vec{c} - \frac{F_c}{c} \vec{\alpha}$$

$$16.7 \quad \mu \dot{\vec{c}} = \vec{r} \times \ddot{\vec{c}} - A r^2 \ddot{\vec{r}} + \left[(A(\vec{r} \cdot \vec{r}) + \frac{F_c}{r^2}) \vec{r} \right]$$

$$A = \frac{\mu c}{r^2} \sin f$$

$$\vec{r} \times \vec{r} = 0$$

$$\vec{r} \times \vec{c} = -\alpha$$

$$\vec{r} \times \vec{\alpha} = r \frac{\vec{c}}{c}$$

$$\alpha = r c$$

$$\vec{c} \times \vec{c} = 0$$

$$\vec{\alpha} \times \vec{c} = c \frac{\vec{r}}{r}$$

$$\dot{\vec{c}} = \vec{r} \times \left(\frac{F_c}{c} \vec{c} + F_r \frac{\vec{r}}{r} + F_a \frac{\vec{\alpha}}{\alpha} \right)$$

$$= -\frac{F_c}{c} \vec{\alpha} + F_a \frac{\vec{c}}{c} r$$

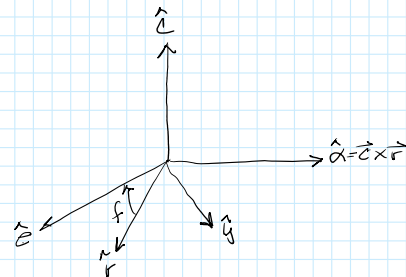
$$\dot{\vec{c}} = -\frac{F_c \vec{\alpha}}{c} + r \frac{F_a \vec{c}}{c}$$

$$\mu \dot{\vec{c}} = \left(\frac{F_c}{c} \vec{c} + F_r \frac{\vec{r}}{r} + F_a \frac{\vec{\alpha}}{\alpha} \right) \times \vec{c} - A r^2 \left(\frac{F_c}{c} \vec{c} + F_r \frac{\vec{r}}{r} + F_a \frac{\vec{\alpha}}{\alpha} \right) + (A F_r r + \frac{F_c}{r^2}) \vec{r}$$

$$= -\frac{F_c}{r} \vec{\alpha} + \frac{c F_a}{r} \vec{r} - A r^2 \frac{F_c}{c} \vec{c} - A r^2 F_r \frac{\vec{r}}{r} - A r^2 \frac{F_a}{\alpha} \vec{\alpha} + (A F_r r + \frac{c F_a}{r^2}) \vec{r}$$

$$= -A r^2 \frac{F_c}{c} \vec{c} - (A r^2 \frac{F_a}{r c} + \frac{F_r}{r}) \vec{\alpha} + \left(\frac{c F_a}{r} - A F_r r + A F_r r + \frac{c F_a}{r^2} \right) \vec{r}$$

$$\mu \dot{\vec{c}} = -A r^2 \frac{F_c}{c} \vec{c} + 2 \frac{c F_a}{r} \vec{r} - (A r \frac{F_a}{c} + \frac{F_r}{r}) \vec{\alpha}$$



Hwk 1 Prob 2

Friday, January 23, 2026 10:20 PM

2) prove identities

Linear momentum:

$$\begin{aligned}
 \vec{F}_{ij} &= \frac{Gm_i m_j}{r_{ij}^3} \vec{r}_{ji} = -\frac{Gm_i m_j}{r_{ji}^3} \vec{r}_{ij} \\
 \sum_{i=1}^N \vec{F}_i &= \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{Gm_i m_j}{r_{ij}^3} \vec{r}_{ji} \\
 &= \sum \left(\frac{m_i m_j}{r_{ij}^3} \vec{r}_{ji} + \frac{m_i m_j}{r_{ji}^3} \vec{r}_{ij} \right) \\
 &= \sum \frac{m_i m_j}{r_{ij}^3} (\vec{r}_{ji} + \vec{r}_{ij}) = 0 \\
 \text{so } \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \vec{r}_{ji} &= 0
 \end{aligned}$$

angular momentum

$$\vec{X} = \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \vec{r}_i \times \vec{r}_{ji} = -\sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \vec{r}_j \times \vec{r}_{ij}$$

$$\begin{aligned}
 2\vec{X} &= \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \left(\frac{m_i m_j}{r_{ij}^3} \vec{r}_i \times \vec{r}_{ji} + \frac{m_i m_j}{r_{ij}^3} \vec{r}_j \times \vec{r}_{ij} \right) \\
 &= \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} (\vec{r}_i \times \vec{r}_{ji} + \vec{r}_j \times \vec{r}_{ij}) \\
 &= \sum_{i=1}^N \sum_{j=1}^N \frac{m_i m_j}{r_{ij}^3} (\vec{r}_i \times (\vec{r}_i - \vec{r}_j) + \vec{r}_j \times (\vec{r}_j - \vec{r}_i)) \\
 &= \sum_{i=1}^N \sum_{j=1}^N \frac{m_i m_j}{r_{ij}^3} (-\vec{r}_{ij} - \vec{r}_{ji}) = 0
 \end{aligned}$$

$$\begin{aligned}
 2\vec{X} &= 0 \\
 \vec{X} &= 0
 \end{aligned}$$

$$\text{so } \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \vec{r}_i \times \vec{r}_{ji} = 0$$

Energy

$$\sum_{i=1}^N \sum_{\substack{j=1 \\ i \neq j}}^N \frac{m_i m_j}{r_{ij}} = 2 \sum_{j=1}^N \sum_{i=1}^N \frac{m_i m_j}{r_{ij}}$$

$$U = -G \sum_{i=1}^N \sum_{\substack{j=1 \\ i \neq j}}^N \frac{m_i m_j}{r_{ij}} = -G \left[2 \sum_{i < j \leq N} \frac{m_i m_j}{r_{ij}} \right]$$

$$U = -G \sum_{i < j \leq N} \frac{m_i m_j}{r_{ij}}$$

total energy

$$T = \frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i$$

$$\frac{dT}{dt} = \frac{1}{2} \sum_{i=1}^N m_i (\dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i + \ddot{\vec{r}}_i \cdot \dot{\vec{r}}_i)$$

$$\frac{dT}{dt} = \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i$$

$$\vec{F}_i = m_i \ddot{\vec{r}}_i = \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_i m_j}{r_{ij}^3} \vec{r}_{ji}$$

$$\frac{dT}{dt} = \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ji}$$

$$\frac{dU}{dt} = -G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{d}{dt} \left(\frac{m_i m_j}{\|\vec{r}_{ij}\|} \right)$$

$$= -G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N m_i m_j \left(\frac{1}{\|\vec{r}_{ij}\|^2} \right) \frac{d \cdot \|\vec{r}_{ij}\|}{dt} \rightarrow \sqrt{\vec{r}_{ij} \cdot \vec{r}_{ij}}$$

$$= +G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N m_i m_j \frac{1}{\|\vec{r}_{ij}\|^2} \left(\frac{1}{2 \sqrt{\vec{r}_{ij} \cdot \vec{r}_{ij}}} \right) \frac{d(\vec{r}_{ij} \cdot \vec{r}_{ij})}{dt}$$

$$= -G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} (\dot{\vec{r}}_i - \dot{\vec{r}}_j) \cdot \vec{r}_{ij}$$

$$\frac{dU}{dt} = -G \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ij}$$

$$\frac{d(T+U)}{dt} = \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ji}$$

$$\frac{d}{dt} \left(\sum_{i=1}^n \sum_{j=1, j \neq i}^n \frac{G m_i m_j}{r_{ij}^3} \vec{r}_{ij} \cdot \vec{v}_i \right) = 0$$

$$\boxed{\frac{d(T+U)}{dt} = 0}$$

$$3) \quad U = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\vec{r}_{ij}\|}$$

$$U = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\alpha \vec{r}_{ij}\|} = -\frac{G}{\alpha} \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\vec{r}_{ij}\|} = \boxed{-G \alpha^{-1} \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\|\vec{r}_{ij}\|} \Rightarrow n = -1}$$

$$\frac{\partial U}{\partial \vec{r}_i} = -G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}^3} \vec{r}_i \cdot \vec{r}_{ij}$$

$$\vec{r}_i \cdot \frac{\partial U}{\partial \vec{r}_i} = G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}^3} \underbrace{\vec{r}_i \cdot \vec{r}_i}_{r_i^2} \cdot \vec{r}_{ij}$$

$$\boxed{\vec{r}_i \cdot \frac{\partial U}{\partial \vec{r}_i} = G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}} \vec{r}_{ij} = -U}$$

$$T = \frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i$$

$$T = \frac{1}{2} \sum_{i=1}^N m_i \alpha \dot{\vec{r}}_i \cdot \alpha \dot{\vec{r}}_i = \boxed{\alpha^2 \left(\frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i \right) \Rightarrow n = 2}$$

$$\frac{\partial T}{\partial \vec{r}_i} = \sum_{i=1}^N m_i \dot{\vec{r}}_i$$

$$\boxed{\vec{r}_i \cdot \frac{\partial T}{\partial \vec{r}_i} = \sum_{i=1}^N m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i = 2T}$$

$$I_p = \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i$$

$$I_p = \sum_{i=1}^N m_i \alpha \vec{r}_i \cdot \alpha \vec{r}_i = \boxed{\alpha^2 r_i \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i \Rightarrow N = 2}$$

$$\frac{\partial I_p}{\partial \vec{r}_i} = 2 \sum_{i=1}^N m_i \vec{r}_i$$

$$\boxed{\vec{r}_i \cdot \frac{\partial I_p}{\partial \vec{r}_i} = 2 \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i = 2I_p}$$

$$I_p U^2 = \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{r}_i \left[G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{r_{ij}} \right]^2$$

$$\boxed{= \sum_{i=1}^N m_i \alpha \vec{r}_i \cdot \alpha \vec{r}_i \left(G \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{\alpha r_{ij}} \right)^2 \Rightarrow n = 0}$$

for homogeneous functions order product of functions equals sum of order $(2 + 2(-1) = 0)$
so order = 0

$$4) \vec{H} = \sum_{i=1}^N m_i \vec{r}_i \times \dot{\vec{r}}_i$$

$$H^2 \leq 2T I_P$$

$$a = \vec{r} \quad b = \dot{\vec{r}}$$

$$\vec{H} = \sum m_i a \times b$$

$$\vec{H}^2 = \sum m_i a \times b \sum m_i a \times b$$

$$= \sum m_i^2 (a \times b) \times (a \times b)$$

$$H^2 = \sum m_i^2 [a^2 b^2 - (a \cdot b)^2]$$

$$H^2 \leq 2T I_P$$

$$T = \frac{1}{2} \sum m_i a^2$$

$$I_P = \sum m_i b^2$$

$$A = 2T$$

$$B = I_P$$

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5)

Lagrange Identity

$$\left(\sum_{k=1}^N a_k^2\right) \left(\sum_{k=1}^N b_k^2\right) - \left(\sum_{k=1}^N a_k b_k\right)^2 = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N (a_i b_j - a_j b_i)^2$$

$$I_P = \sum_{i=1}^N m_i r_i^2 = \sum_{i=1}^N m_i (\vec{r}_i \cdot \vec{r}_i)$$

$$\vec{r}_{ij} = \vec{r}_i - \vec{r}_j$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (\vec{r}_i - \vec{r}_j)^2$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (r_i^2 - 2\vec{r}_i \cdot \vec{r}_j + r_j^2)$$

$$\frac{1}{2M} \left(\sum_{i=1}^N m_i r_i^2 \sum_{j=1}^N m_j - 2 \sum_{i=1}^N \sum_{j=1}^N \vec{r}_i \cdot \vec{r}_j + \sum_{j=1}^N m_j \sum_{i=1}^N m_i r_j^2 \right)$$

$$\underbrace{\frac{1}{2} \sum_{i=1}^N m_i r_i^2}_{I_P} - \underbrace{\frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i \cdot \sum_{j=1}^N m_j \vec{r}_j}_{M \vec{R}_c \cdot \vec{R}_c} + \underbrace{\frac{1}{2M} \sum_{i=1}^N m_i \sum_{j=1}^N m_j r_j^2}_{I_P}$$

$$\frac{1}{2} I_P + \frac{1}{2} I_P - M \vec{R}_c \cdot \vec{R}_c$$

$$\frac{1}{2M} \sum_{i,j} m_i m_j \vec{r}_i \cdot \vec{r}_j = I_P - M \vec{R}_c \cdot \vec{R}_c$$

$$I_P = \frac{1}{2M} \sum_{i,j} m_i m_j \vec{r}_i \cdot \vec{r}_j + M \vec{R}_c \cdot \vec{R}_c$$

$$T = \frac{1}{2} \sum_{i=1}^N m_i v_i^2$$

$$\frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j v_{ij}^2 = \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (v_i - v_j)^2$$

$$= \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j (v_i^2 - 2\vec{v}_i \cdot \vec{v}_j + v_j^2)$$

$$= \frac{1}{2M} \left(\sum_{i=1}^N \sum_{j=1}^N m_i m_j v_i^2 - \sum_{i,j} 2m_i m_j \vec{v}_i \cdot \vec{v}_j + \sum_{i,j} m_i m_j v_j^2 \right)$$

$$= \frac{1}{2M} \left(\underbrace{\sum_{i=1}^N m_i v_i^2 \sum_{j=1}^N m_j}_{I_P} - \sum_{i=1}^N m_i \vec{v}_i \cdot \sum_{j=1}^N m_j \vec{v}_j + \underbrace{\sum_{i=1}^N m_i \sum_{j=1}^N m_j v_j^2}_{I_P} \right)$$

$$= \frac{1}{2M} \left(\underbrace{\sum_{i=1}^N m_i v_i^2}_T \underbrace{\sum_{j=1}^N m_j}_M \underbrace{\sum_{i=1}^N m_i \vec{v}_i \cdot \sum_{j=1}^N m_j \vec{v}_j}_{\vec{v}_c^2} + \underbrace{\sum_{i=1}^N m_i}_M \underbrace{\sum_{j=1}^N m_j v_j^2}_T \right)$$

$$= \frac{1}{2} (2T - M \vec{v}_c \cdot \vec{v}_c)$$

$$T = \frac{1}{2M} \sum_{i=1}^N \sum_{j=1}^N m_i m_j v_{ij}^2 + \frac{1}{2} M \vec{v}_c \cdot \vec{v}_c$$

6) show:

$$\sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N f_{ij} = \frac{1}{2} \sum_{1 \leq i < j \leq N} f_{ij} ; f_{ii} = 0$$

$$\sum_{i=1}^N m_i \vec{r}_i \cdot \vec{v}_i = \frac{1}{M} \sum_{i=1}^N \sum_{j=1+i}^N m_i m_j \vec{r}_{ij} \cdot \vec{v}_{ij}$$

$$\vec{r}_i \cdot \vec{v}_i \text{ at } i=j \quad \vec{r}_{ij} \cdot \vec{v}_{ij} = 0$$

$$\frac{1}{M} \left[\frac{1}{2} \sum_{1 \leq i < j \leq N} m_i m_j (\vec{r}_j - \vec{r}_i) \cdot (\vec{v}_j - \vec{v}_i) \right]$$

$$\frac{1}{2M} \sum_{1 \leq i < j \leq N} m_i m_j [\vec{r}_j \cdot \vec{v}_j - \vec{r}_j \cdot \vec{v}_i - \vec{r}_i \cdot \vec{v}_j + \vec{r}_i \cdot \vec{v}_i]$$

$$= \frac{1}{2M} \left[\left(\sum_{j=1}^N m_j \vec{r}_j \cdot \vec{v}_j \right) \left(\sum_{i=1}^N m_i \right) - \underbrace{\left(\sum_{j=1}^N m_j \vec{r}_j \right) \cdot \left(\sum_{i=1}^N m_i \vec{v}_i \right)}_{M \vec{R}_c \cdot M \vec{V}} - \underbrace{\left(\sum_{i=1}^N m_i \vec{r}_i \right) \cdot \left(\sum_{j=1}^N m_j \vec{v}_j \right)}_{M \vec{R}_c \cdot M \vec{V}} \right]$$

$$\sum_{j=1}^N m_j \vec{r}_j \cdot \vec{v}_j = \sum m_i \vec{r}_i \cdot \vec{v}_i$$

$$= \frac{1}{2M} (2M \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{v}_i - 2M \vec{R}_c \cdot M \vec{V})$$

$$= \sum_{i=1}^N m_i \vec{r}_i \cdot \vec{v}_i - M \vec{R}_c \cdot \vec{V} \quad \checkmark \quad \text{at COM } \vec{R}_c = 0$$

$$\boxed{\sum_{i=1}^N m_i \vec{r}_i \cdot \vec{v}_i = \frac{1}{M} \sum_{i=1}^N \sum_{j=1+i}^N m_i m_j \vec{r}_{ij} \cdot \vec{v}_{ij}}$$

show:

$$\sum_{i=1}^N m_i [r_i^2 \bar{U} - \vec{r}_i \cdot \vec{r}_i] = \frac{1}{M} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N m_i m_j [r_{ij}^2 \bar{U} - \vec{r}_{ij} \cdot \vec{r}_{ij}]$$

$$= \frac{1}{2M} \sum_{i=1}^N \sum_{j=1+i}^N m_i m_j [r_{ij}^2 \bar{U} - \vec{r}_{ij} \cdot \vec{r}_{ij}]$$

$$= \frac{1}{2M} \sum_{i=1}^N \sum_{j=1+i}^N m_i m_j [(r_j^2 - 2\vec{r}_i \cdot \vec{r}_j + r_i^2) \bar{U} - (\vec{r}_j \cdot \vec{r}_j - \vec{r}_j \cdot \vec{r}_i - \vec{r}_i \cdot \vec{r}_j + \vec{r}_i \cdot \vec{r}_i)]$$

$$= \frac{1}{2M} \sum_{i=1}^N \sum_{j=1+i}^N \underbrace{m_i m_j}_{\frac{1}{M}} \left\{ \underbrace{r_i^2 \bar{U} - \vec{r}_i \cdot \vec{r}_i}_{2\vec{r}_i^2 \bar{U} - \vec{r}_i \cdot \vec{r}_i} + [r_j^2 \bar{U} - \vec{r}_j \cdot \vec{r}_j] - 2(\vec{r}_i \cdot \vec{r}_j) \bar{U} + \vec{r}_i \cdot \vec{r}_j + \vec{r}_j \cdot \vec{r}_i \right\}$$

$$= \sum_{i=1}^N r_i^2 \bar{U} - \vec{r}_i \cdot \vec{r}_i + \frac{1}{M} \sum_{i=1}^N \sum_{j=1+i}^N m_i m_j [-2(\vec{r}_i \cdot \vec{r}_j) \bar{U} + \vec{r}_i \cdot \vec{r}_j + \vec{r}_j \cdot \vec{r}_i]$$

$$= \sum_{i=1}^N r_i^2 \bar{U} - \vec{r}_i \cdot \vec{r}_i - \frac{1}{M} \left[\left(\sum_{i=1}^N m_i \vec{r}_i \right) \cdot \left(\sum_{j=1+i}^N m_j \vec{r}_j \right) \bar{U} + \sum_{i=1}^N m_i \vec{r}_i \cdot \sum_{j=1+i}^N m_j \vec{r}_j \right] + \sum_{j=1}^N m_j \vec{r}_j \cdot \sum_{i=1}^N m_i \vec{r}_i$$

$$= \sum_{i=1}^N r_i^2 \bar{U} - \vec{r}_i \cdot \vec{r}_i - \frac{1}{M} \underbrace{M \vec{R}_c \cdot M \vec{R}_c \bar{U}}_0 + 2 \underbrace{M \vec{R}_c \cdot \vec{R}_c}_0$$

$$\text{at COM } \vec{R}_c = 0$$

$$\boxed{\sum_{i=1}^N r_i^2 \bar{U} - \vec{r}_i \cdot \vec{r}_i = \frac{1}{M} \sum_{i=1}^N \sum_{j=1+i}^N m_i m_j [r_{ij}^2 \bar{U} - \vec{r}_{ij} \cdot \vec{r}_{ij}]}$$

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7) $M_1 = m_1$ $M_2 = m_1 + m_2$ $M_3 = m_1 + m_2 + m_3$
 $\vec{R}_1 = \vec{r}_2 - \vec{r}_1$ $\vec{r} = \vec{R}_2 = \vec{r}_3 - \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$ $\vec{R}_{cm} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3}{M}$
 $T = \frac{1}{2} \sum_{i=1}^3 m_i \dot{\vec{r}}_i^2$ define $M_i = \sum_{j=1}^i m_j$; $\vec{R}_N \equiv$ system center of mass

$$T = \frac{1}{2} \sum_{i=1}^{N-1} M_{i-1} \frac{M_{i-1} m_i}{M_i} \dot{\vec{R}}_i \cdot \dot{\vec{R}}_i + \frac{1}{2} M_N \dot{\vec{R}}_N \cdot \dot{\vec{R}}_N$$

$$T = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} \dot{\vec{R}} \cdot \dot{\vec{R}} + \frac{1}{2} \frac{m_3 (m_1 + m_2)}{m_1 + m_2 + m_3} \dot{\vec{r}} \cdot \dot{\vec{r}} \quad \text{KE of COM}$$

$$\mu_1 = \frac{m_1 m_2}{M_2} \quad \mu_2 = \frac{M_2 m_3}{M_3}$$

COM: $\vec{R}_1 = \vec{r}_2 - \vec{R}_1^c = \vec{r}_2 - \vec{r}_1$
 $\vec{R}_2 = \vec{r}_3 - \vec{R}_2^c = \vec{r}_3 - \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$

$$\vec{r}_3 = \vec{R}_2^c + \vec{r}$$

$$\vec{r}_2 = \vec{R}_2^c + \frac{m_1}{M_2} \vec{R}_1$$

$$\vec{r}_1 = \vec{R}_2^c - \frac{m_2}{M_2} \vec{R}_1$$

$$U = - \frac{G m_1 m_2}{|\vec{R}_1|} - \frac{G m_1 m_3}{|\vec{r} + \frac{m_2}{M_2} \vec{R}|} - \frac{G m_2 m_3}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|}$$

$$L = T - U$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\vec{R}}_j} = - \frac{\partial U}{\partial \vec{R}_j}$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\vec{R}}} = \frac{m_1 m_2}{M_2} \ddot{\vec{R}}$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\vec{r}}} = \frac{m_3 M_2}{M_3} \ddot{\vec{r}}$$

$$- \frac{\partial}{\partial \vec{R}} \left(- \frac{G m_1 m_2}{R} \right) = - G m_1 m_2 \frac{\vec{R}}{R^3}$$

$$- \frac{\partial}{\partial \vec{R}} \left(- \frac{G m_1 m_3}{\vec{r} + \frac{m_2}{M_2} \vec{R}} \right) = - G m_1 m_3 \frac{m_2}{M_2} \frac{(\vec{r} + \frac{m_2}{M_2} \vec{R})}{|\vec{r} + \frac{m_2}{M_2} \vec{R}|^3}$$

$$- \frac{\partial}{\partial \vec{R}} \left(- \frac{G m_2 m_3}{\vec{r} - \frac{m_1}{M_2} \vec{R}} \right) = - G m_2 m_3 \frac{(-m_1)}{M_2} \frac{(\vec{r} - \frac{m_1}{M_2} \vec{R})}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|^3} = \frac{G m_1 m_2 m_3}{M_2} \frac{(\vec{r} - \frac{m_1}{M_2} \vec{R})}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|^3}$$

$$- \frac{\partial U}{\partial \vec{R}} = \left[- \frac{G m_1 m_2}{R^3} \vec{R} - \frac{G m_1 m_2 m_3}{M_2} \frac{(\vec{r} + \frac{m_2}{M_2} \vec{R})}{|\vec{r} + \frac{m_2}{M_2} \vec{R}|^3} - \frac{G m_2 m_3}{M_2} \frac{(\vec{r} - \frac{m_1}{M_2} \vec{R})}{|\vec{r} - \frac{m_1}{M_2} \vec{R}|^3} \right] = \frac{m_1 m_2}{M_2} \ddot{\vec{R}}$$

$$-\frac{\partial U}{\partial \vec{R}} = \left[-\frac{Gm_1m_2}{R^3} \vec{R} - \frac{Gm_1m_2m_3}{M_2} \left[\frac{\vec{S} + \frac{m_3}{M_2} \vec{R}}{|\vec{S} + \frac{m_3}{M_2} \vec{R}|^3} - \frac{\vec{S} - \frac{m_1}{M_2} \vec{R}}{|\vec{S} - \frac{m_1}{M_2} \vec{R}|^3} \right] \right] = \frac{m_1m_2}{M_2} \vec{R}$$

$$-\frac{\partial U}{\partial \vec{S}} \left(-\frac{Gm_1m_3}{|\vec{S} + \frac{m_3}{M_2} \vec{R}|^3} \right) = -Gm_1m_3 \frac{\vec{S} + \frac{m_3}{M_2} \vec{R}}{|\vec{S} + \frac{m_3}{M_2} \vec{R}|^3}$$

$$-\frac{\partial U}{\partial \vec{S}} \left(-\frac{Gm_2m_3}{|\vec{S} - \frac{m_1}{M_2} \vec{R}|^3} \right) = -Gm_2m_3 \frac{\vec{S} - \frac{m_1}{M_2} \vec{R}}{|\vec{S} - \frac{m_1}{M_2} \vec{R}|^3}$$

$$\frac{\partial U}{\partial \vec{S}} = \frac{m_3M_2}{M_3} \ddot{\vec{S}} = -Gm_1m_3 \frac{\vec{S} + \frac{m_3}{M_2} \vec{R}}{|\vec{S} + \frac{m_3}{M_2} \vec{R}|^3} - Gm_2m_3 \frac{\vec{S} - \frac{m_1}{M_2} \vec{R}}{|\vec{S} - \frac{m_1}{M_2} \vec{R}|^3}$$

$N=4$ Jacobi

$$M_4 = m_1 + m_2 + m_3 + m_4$$

$$\vec{R}_3 = \vec{r}_4 + \vec{R}_3^c = \vec{r}_4 - \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3}{M_3}$$

$N=5$ Jacobi

$$M_5 = M_4 + m_5$$

$$\vec{R}_4 = \vec{r}_5 + \vec{R}_4^c = \vec{r}_5 - \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3 + m_4\vec{r}_4}{M_4}$$